London Business School

Mergers, MBOs & Other Corporate Reorganizations

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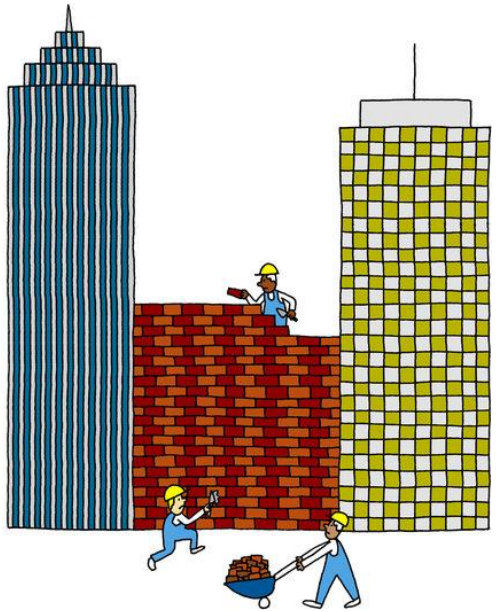
Session 2: Value creation in M&A: the value of control

Synergies: Combined firm is more valuable than the merging firms separately

- Revenue synergies
- Cost synergies
- Tax gains
- Financing gains

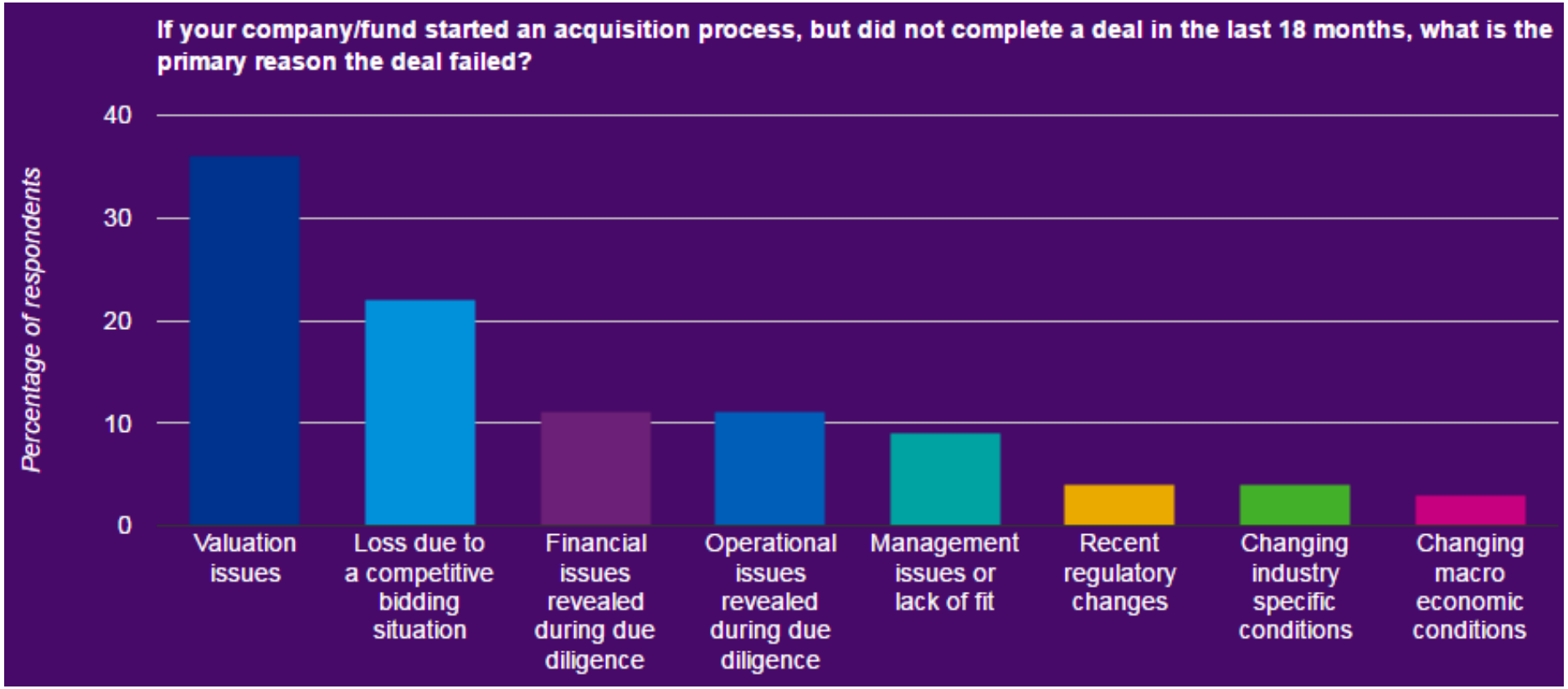
Agency

- Market for corporate control
- Empire building
- Overconfidence



Source: Jordan Awan, New York Times

Overview of class



Source: KPMG

Overview of class

- What is the “Value of Control”?
- Estimates of the value of control
- What determines the value of control?

- Calculating merger value creation from stock market data
- Merger arbitrage

Why pay a premium for the target firm?

- Answer from last class: synergies
- This class: *Control*

- **Definition of control**: ability to decide on what to do with the assets
  - Different from ownership, which is rights to cash flows

The value of control

- An acquisition of a **majority stake** (usually 50% or more) enables the buyer to control the target firm.
  - Compared to a minority shareholder, the majority owner is in a position to influence the firms' strategy and thereby its performance.

- This option to influence the way the firm is run should be valuable for the buyer, giving rise to a **control premium**, or, equivalently, a **minority discount**.

- Note: a control premium is **distinct from synergies**.
  - Control premia exist if value can created by changing the strategy of the target as a stand-alone firm.
  - Of course, control is required to obtain synergies as well. But the two are different concepts.

Why is there a control premium?
Theory #1: The prospect of better management

$$\text{Firm value} = \begin{bmatrix}\text{Status Quo}\\ \text{value}\end{bmatrix} + \underbrace{\text{Prob}(\text{Change in control}) \times \begin{bmatrix}\textbf{Value increase}\\ \textbf{under new}\\ \textbf{control}\end{bmatrix}}_{\text{Control premium}}$$

- "Value increase under new control": improvement wrt "Status quo value" (Standalone value)
- There is no "one-size-fits-all" control premium in this theory:
  - For well-run firms there is nothing to improve upon
  - For takeover-protected firms likelihood of change-in-control is small
  - In both cases, control premium should be (close to) zero

Why is there a control premium?
Theory #2: Benefits of control

- The buyer can generate value for all owners by increasing future FCF…
- … but it may also generate value for him/herself
  - When this *is not* at the expense of cash flow rights: **amenity potential**
    - Example: prestige from the company sharing your family name
  - When this *is* at the expense of cash flow rights: **private benefit of control**
    - Example: using the company to employ incompetent family members
  - This may boost the value of control, even if not included in DCF
  - It depends on the identity of the buyer

Why is there a control premium?
Theory #2: Benefits of control

- Evidence:
  - Price difference between **voting** and **non-voting** shares used in **dual-class firms**
  - **Acquisition of controlling blocks of shares** vs minority stakes

- Divergence between cash-flow- and voting rights often leads to higher executive compensation and worse acquisitions (Masulis, Xie and Wang, JF 2009)

- Internationally premiums are a function of **law enforcement** and **investor protection** provided by the legal environment to minority shareholders

The value of control

Market	Majority deals		Minority deals		Difference	
	N	Premium	N	Premium	N	Premium
U.S.	3883	37.4%	1675	8.5%	1498	26.9%
Internat.	1678	32.6%	1025	9.8%	328	14.9%

Premium for acquisitions in different markets, 1986-2000.
Source: Hanouna, Sarin, Shapiro (2001, WP)

The value of control

	% Premium paid for **shares with voting right** attached (Nenova, 2000)	% Premium paid to acquire a **controlling block** of equity (Dyck and Zingales, 2003)
United States	2%	1%
United Kingdom	-	1
Germany	9.5	10
France	28	2
Italy	29	37
Spain	-	4
Portugal	-	20
Sweden	1.0	7
Poland	-	13
Czech Republic	-	58

Valuing control

- Consider a firm with two share classes, A & B
- Assume the A and B shares have *identical cash flow rights*, but A has one vote and B has zero votes
- Let:
  - P_A = price of A
  - P_B = price of B
- Then the (per share) voting premium is the difference $P_A - P_B$

Valuing control

- What if share classes A (reduced voting rights) and B (enhanced voting rights) also have *different cash-flow rights*?

- To calculate the value if control, we first need to adjust share prices for cashflow rights, to make them comparable on that dimension

Example: Berkshire Hathaway

- Two classes of shares A & B
  - B shares have 1/1,500 cash-flow rights, and 1/10,000 voting rights for every A share

- First multiply B by 1,500 to equalize cash-flow rights across A and B shares
- This "adjusted" share *B'* now has the same cash-flow rights as share A

- B' has 1,500/10,000=0.15 votes per A vote

Example: Berkshire Hathaway

On March 29, 2021

- P_A = \$391,540
- P_B = \$258.82

Berkshire Hathaway Inc. (BRK-A)
NYSE - Nasdaq Real Time Price. Currency in USD

Add to watchlist

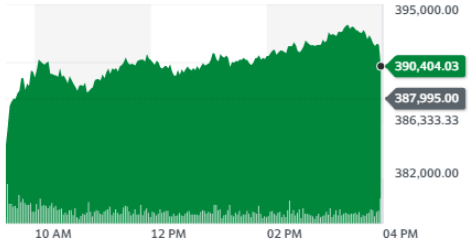
391,540.00 +3,545.00 (+0.91%)
At close: 4:02PM EDT

390,000.00 -1,540.00 (-0.39%)
After hours: 4:53PM EDT

Summary | Chart | Conversations | Statistics | Historical Data | Profile | Financials | Analysis | Options | Holders | Sustainability

Previous Close	387,995.00	Market Cap	596.425B
Open	384,053.00	Beta (5Y Monthly)	0.89
Bid	388,916.00 x 800	PE Ratio (TTM)	14.68
Ask	399,405.00 x 800	EPS (TTM)	26,667.81
Day's Range	383,922.50 - 393,449.50	Earnings Date	N/A
52 Week Range	250,926.00 - 407,750.00	Forward Dividend & Yield	N/A (N/A)
Volume	2,012	Ex-Dividend Date	N/A
Avg. Volume	12	1y Target Est	425,000.00

1D 5D 1M 6M YTD 1Y 5Y Max | Full screen



Trade prices are not sourced from all markets

Berkshire Hathaway Inc. (BRK-B)
NYSE - Nasdaq Real Time Price. Currency in USD

Add to watchlist

258.82 +2.05 (+0.80%)
At close: 4:02PM EDT

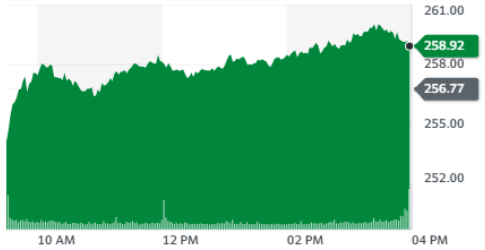
258.90 +0.08 (0.03%)
After hours: 5:18PM EDT

Advertisement
Start Trading >>
Plus500 76.4% of retail CFD accounts lose money

Summary | Chart | Conversations | Statistics | Historical Data | Profile | Financials | Analysis | Options | Holders | Sustainability

Previous Close	256.77	Market Cap	595.744B
Open	253.93	Beta (5Y Monthly)	0.89
Bid	258.52 x 1100	PE Ratio (TTM)	0.01
Ask	259.51 x 800	EPS (TTM)	26,667.81
Day's Range	253.35 - 260.04	Earnings Date	Apr 29, 2021 - May 03, 2021
52 Week Range	167.04 - 267.50	Forward Dividend & Yield	N/A (N/A)
Volume	4,855,212	Ex-Dividend Date	N/A
Avg. Volume	5,264,322	1y Target Est	275.00

1D 5D 1M 6M YTD 1Y 5Y Max | Full screen



Trade prices are not sourced from all markets

Example: Berkshire Hathaway

- Hence: $\boldsymbol{P_{B'}}$ = \$258.82 x 1,500 = \$388,230

- In other words, if I buy 1,500 B shares, I am entitled to the same cash-flow rights as one A share

- The difference between the price of share A (P_A = \$391,540) and the adjusted price of share B' ($P_{B'}$ = \$388,230) is \$3,310
- This difference reflects the **voting premium per share**, with the complication that share B' has 0.15 voting rights (not zero)
- \$3,310 reflects a 0.85 difference in voting rights, meaning that we can compute the premium for a one-unit difference in voting rights as:

$$\frac{\$3{,}310}{0.85} = \$3{,}894$$

Example: Berkshire Hathaway

- The **voting premium expressed in percentage terms** is

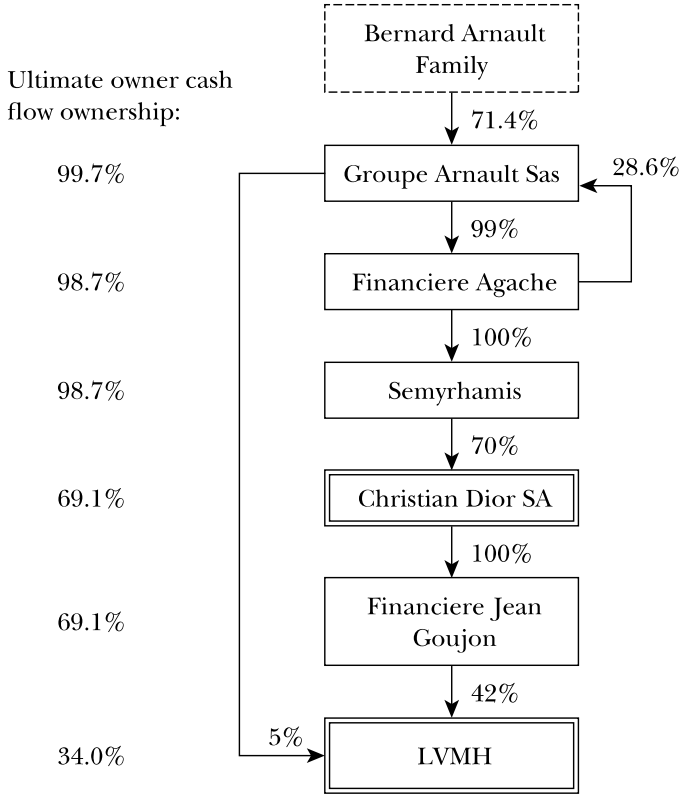
$$\frac{\$391{,}540}{\$391{,}540-\$3{,}894}-1=1\%$$

- This is the premium in value of a voting share A (with one full voting right) relative to the value of the same share less the value of a full voting right

Separation of ownership and control: LVMH

Bernard Arnault, CEO and Chairman of Board, controls 47% of voting rights. But only 34% of cash flow rights. Does it matter? Why?

Figure 1
Ownership Structure of LVMH in 2005



Why disproportionate voting rights may matter?

- Suppose Bernard Arnault sets up and owns 100% of the shares of a new corporation, BAX, that is to be a key supplier to LVMH.
- Suppose the “arms-length zero economic profit price” of some input supplied from BAX to LVMH is $10m.
- Suppose BAX instead offers it to LVMH for $30m.
- Using his superior voting power, Arnault might get the transaction to go thru, making a profit of…

  Profit= 100% x 20 – 34% x 20=66% x 20=$13.2

Valuing control using block premium

- We measured the value of control using dual-class shares.
- An **alternative way of measuring the value of control** (using shares that have one vote each) is to compute the premium that is paid for large **blocks of shares** vs. buying a single share

- What is the idea?
  - Buying a large stake in a firm, say 55%, gives me power to steer the firm
  - When I buy a single share in a corporation, I have little say in how the firm will operate
  - Therefore, if large blocks of shares trade at a premium, this can be seen as an indication of the value of control.

Understanding the relationship

- If there is a block premium, then there must be a discount for the shares that are not part of the block.
- That is, small minority stakes sell at a large discount if there is a large block premium.
- But the fact that minority stakes sell at a low price creates a strong *disincentive* for a controlling shareholder to raise equity capital via the sale of minority equity stakes.
- In some countries, e.g. Mexico, company growth is stifled by inability of corporations to convince the market that they will treat minority shareholders fairly.

The value effects of large shareholders

- *Does this mean that large shareholders are necessarily a bad thing?*

- Two types of conflicts of interests:
  - Large shareholders vs. minority shareholders
  - Shareholders vs. management

- What is the role of concentrated ownership?
  - With concentrated ownership, the problem is that controlling shareholders may exploit minority shareholders.

- What if ownership is dispersed?

The economics of dispersed ownership

- Suppose the firm has very dispersed ownership
- Monitoring the CEO costs a lot of time/money
- Will any one shareholder agree to monitor?
  - No! **Free-rider problem**. Full cost is borne by one shareholder, but all shareholders enjoy the benefits.
- What if cost is divided among shareholders?
  - Issue becomes: who will monitor the monitor?

Role of the board

- **Boards** should protect shareholders from agency problems by monitoring the management team
- In reality, boards also differ in their quality
- Gompers et al. (2003) develop an index of corporate governance for U.S. firms. They find that poorly-governed firms engage in too much M&A activity.
- Why don’t shareholders stop this? One reason: shareholders can always “vote with their feet.”

Examples of “agency” issues

- The Quiet Life
  - Management may avoid necessary but painful decisions (e.g., cost-cutting, layoffs, divestitures, shrinking, etc)
- Entrenchment
  - Management may take moves to secure their position at the expense of shareholders (“let’s put Mom on the board”)
- Empire building
  - Management’s tendency to increase firm size, e.g., excessive M&A instead of payouts

Stock market reactions as indicators of merger gains

- So far, we used DCF and Multiples to calculate the value of synergies and control
- Alternative: infer the merger synergy from market prices and statistical model.
- Let’s see how this works …

Statistical approach: Basic idea

- Market Capitalization (Market Cap): Common Equity Price x Shares Outstanding
- Equity prices are forward looking
- Therefore, the market cap of target and acquiring company contains information about market view regarding:
  - merger probability and
  - merger synergies and improvements

Basic idea

Pre-acquisition (announcement)

- Target January 1 = £2 x 100m=£200m
- Acquirer January 1 = £4 x 100m=£400m
- Total January 1=200+400=£600m

Post-acquisition (announcement)

- Target May 1 = £3 x 100m=£300m
- Acquirer May 1 = £5 x 100m=£500m
- Total May 1=300+500=£800m

- Synergy Estimate=800-600=200

- *Problem*: Market cap changes due to overall stock market movements.
- *Solution*: Remove effects by subtracting “normal” returns of stocks

Do takeovers create value? A stock price approach: the CAR

- Idea: how much does the price of target *i* (or acquirer) change due to merger announcement (at date t_0)?
  - The price change is called a CAR: Cumulative Abnormal Return
  - Should reflect value from the merger for target (acquirer) shareholders

- Assumptions:
  1. Markets are efficient
  2. We can estimate how stock prices would evolve in normal times / absent merger announcement (in practice: pre-merger announcement)

How do you estimate a CAR?

1. Define a period over which traders buy and sell the stock based on their analysis of the merger value, from $t_0 - \tau$ to $t_0 + \tau$

2. Estimate a normal or benchmark return for stock *i*
   - For example, assume it follows CAPM: $R_{Benchmark,t} = R_{f,t} + \beta_i * R_{m,t}$
   - $R_{m,t}$ is the return on the market index in excess of risk-free rate R_f

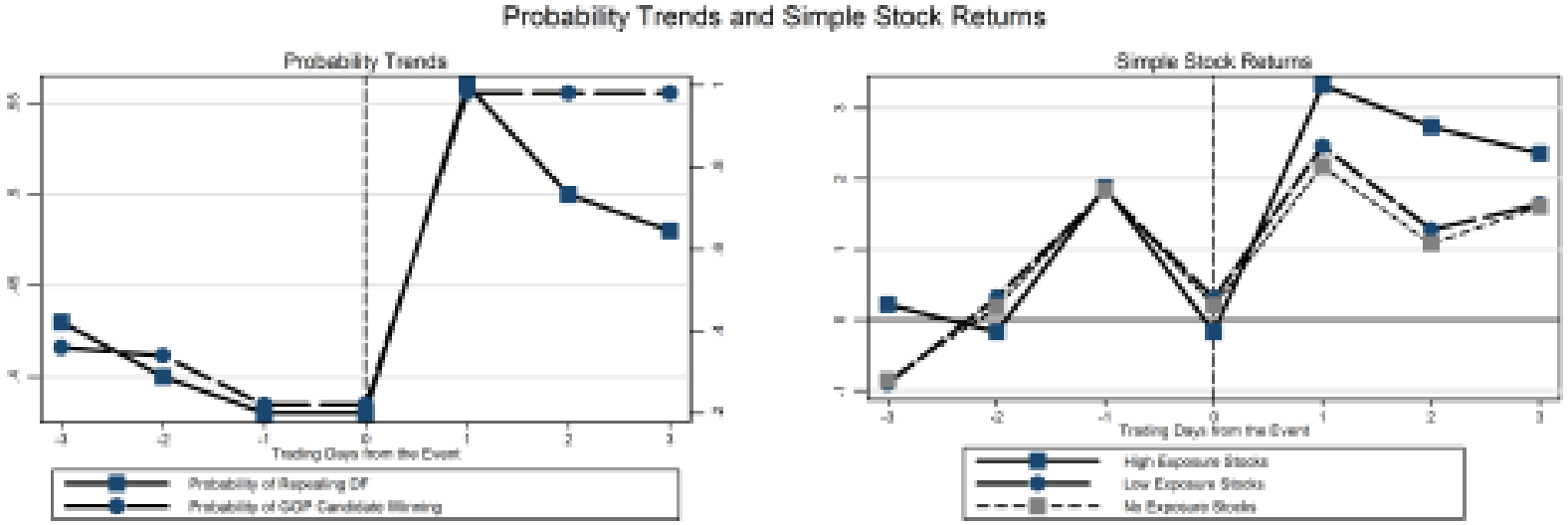
You may then estimate:

$$CAR_i = \sum_{t=t_0-\tau}^{t_0+\tau} \left(R_{i,t} - R_{Benchmark,t}\right)$$

How do you estimate a CAR?

Figure 10: Election and Stock Returns

The panel on the left shows the probability of the Dodd-Frank Act being repealed, the probability of the Republican candidate winning the 2016 US Presidential election between Nov. 3 and Nov. 11 2016. The short dashed line shows the probability of the Republican candidate winning the 2016 election, while the long line shows the probability of Dodd-Frank being repealed by the end of the year. The panel on the right shows stock returns of financial firms by exposure to the Dodd-Frank Act. Exposure to the Dodd-Frank Act is defined by the weighted sum of mentions of Dodd-Frank in the 10-K files from 2008 to 2015. The short dashed line shows firms with no Dodd-Frank exposure, the medium dashed lines shows firms with moderate Dodd-Frank exposure, while the long dashed lines show firms with high Dodd-Frank Act exposure. The dashed vertical line shows the day after the US Presidential Election on Nov. 8. Prediction market data comes from PredictWise. Stock return data comes from CRSP.

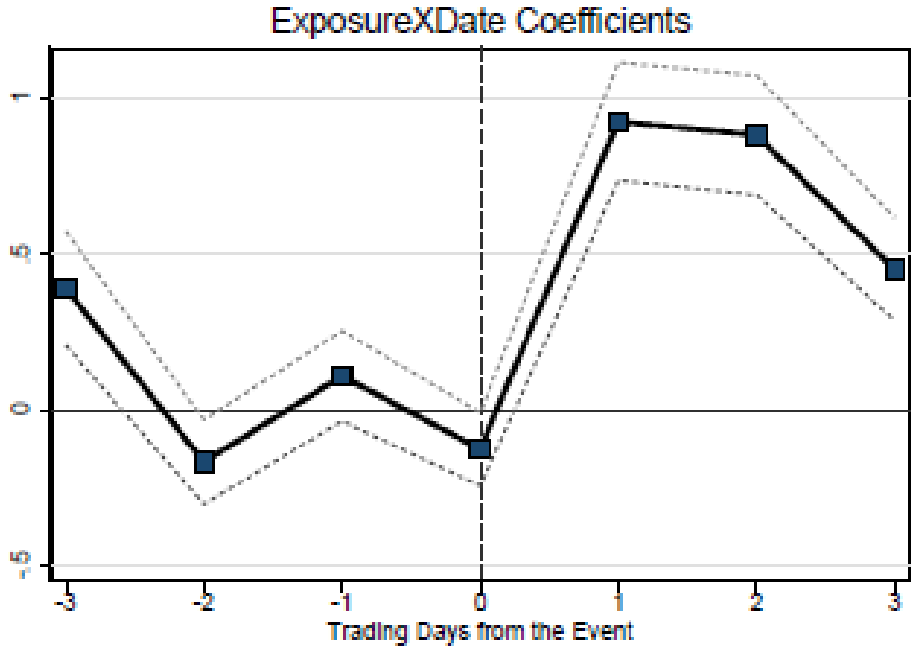


Source: Gong and Yannelis (2017)

How do you estimate a CAR?

Figure 11: Election Abnormal Returns and Event Study

This figure shows difference in difference coefficient resulting from exposure to Dodd-Frank interacted with each time peril from Nov.3 to Nov. 11. The solid line shows point estimates, and the dashed lines show a 95% confidence interval. The panel on the left shows differences from the CAR difference in difference specification. The dependent variable is cumulative abnormal returns in percentage points, at day 0, 1 and 10. The abnormal return is estimated using CAPM, with a pre-event window of 250 trading days, ending at 30 days before the event. The panel on the right shows estimates from the abnormal returns difference in difference specification. Standard errors are clustered at the firm level.



Source: Gong and Yannelis (2017)

How big are target CARs?

	N	Mean	Median	p25	p75	S.D.	Min	Max
Full sample	4,698	0.215**	0.170**	0.058	0.320	0.257	-0.992	2.998
Cash merger	1,293	0.302**	0.246**	0.117	0.418	0.299	-0.896	2.574
Stock merger	1,814	0.174**	0.138**	0.036	0.274	0.222	-0.659	2.998
Mixed	660	0.208**	0.172**	0.075	0.305	0.211	-0.992	1.381
Unknown	931	0.181**	0.134**	0.027	0.284	0.256	-0.567	2.364

Listed US targets, [-1,1] event window

Source: Malmendier (2018)

Target CARs: What can we conclude?

1. M&A typically cause a large wealth increase for target shareholders

2. The wealth increase larger for cash deals than stock deals, why?
   - Overvalued bidders have an incentive to use stock
     - Hence: long-term gains low for target shareholders
   - Paying cash for something reveals it has a lot of fundamental value
     - “Signalling” effect of cash payment
   - Stock deals are more often *friendly deals*
     - Target’s board may not extract a high price from the bidder

How big are bidder CARs?

	N	Mean	Median	p25	p75	S.D.	Min	Max
Full sample	70,575	0.011**	0.003**	-0.018	0.029	0.095	-0.669	6.006
Public targets	6,960	-0.003**	-0.004**	-0.032	0.022	0.086	-0.514	2.141
U.S. public targets (1)	4,687	-0.009**	-0.007**	-0.038	0.019	0.083	-0.514	2.141
Private targets	41,966	0.012**	0.003**	-0.017	0.029	0.100	-0.666	6.006
Other and unknown	21,649	0.014**	0.004**	-0.015	0.031	0.086	-0.669	3.942
Cash merger	9,886	0.013**	0.005**	-0.015	0.031	0.068	-0.460	0.876
Stock merger	6,782	0.014**	-0.001*	-0.031	0.036	0.141	-0.516	4.496
Mixed	4,551	0.020**	0.007**	-0.026	0.053	0.114	-0.551	3.942
Unknown	49,356	0.010**	0.002**	-0.016	0.026	0.089	-0.669	6.006

Listed US acquirers, [-1,1] event window
Source: Malmendier (2018)

Bidder CARs: What can we conclude?

1. At best, bidder shareholders have very small gains from M&A
   - In fact, their wealth often decreases
2. The median wealth decrease is larger for stock deals. Why?
   - Overvalued stock allows bidder's management to indulge in negative NPV transactions (agency cost of management)
   - If uncertain about the value of the merger, bidders use stock…
   - … as then target shareholders share future gains and losses w/ bidder
   - … so paying in stock signals that the deal may be NPV negative
3. But there are also many value-destroying cash deals. Why?
   - In cash deals, bidders are taking all the synergy risks: overoptimistic bidders are likely to (over-)pay in cash

How Big Are Combined CARs?

$$\text{Combined CAR} = \frac{\text{Target CAR} \times \text{Target MV} + \text{Bidder CAR} \times \text{Bidder MV}}{\text{Target MV} + \text{Bidder MV}}$$

Paper	Period	# targets / # bidders	Window	CAR (%)		
				Target	Bidder	Combined
Bradley et al. (1988)	1963-84	236 / 236	[–5,+5]	31.8	0.97	7.43
Kaplan and Weisbach (1992)	1971-82	209 / 271	[–5,+5]	26.9	–1.49	3.74
Servaes (1991)	1972-87	704 / 384	[–1,R]	23.6	–1.07	3.66
Mulherin and Boone (2000)	1990-99	281 / 281	[–1,+1]	20.2	–0.37	3.56
Andrade et al. (2001)	1973-98	3688 / 3688	[–1, +1]	16.0	–0.7	1.8
Moeller et al. (2004)	1980-2001	12023 / 2645	[–1,+1]	NA	NA	1.4
Harford et al. (2012)	1990-2005	3040 / 3040	[–2, +2]	NA	NA	0.9

Source: Author update of Weston, Mitchell and Mulherin (2004)

Combined CARs: What can we conclude?

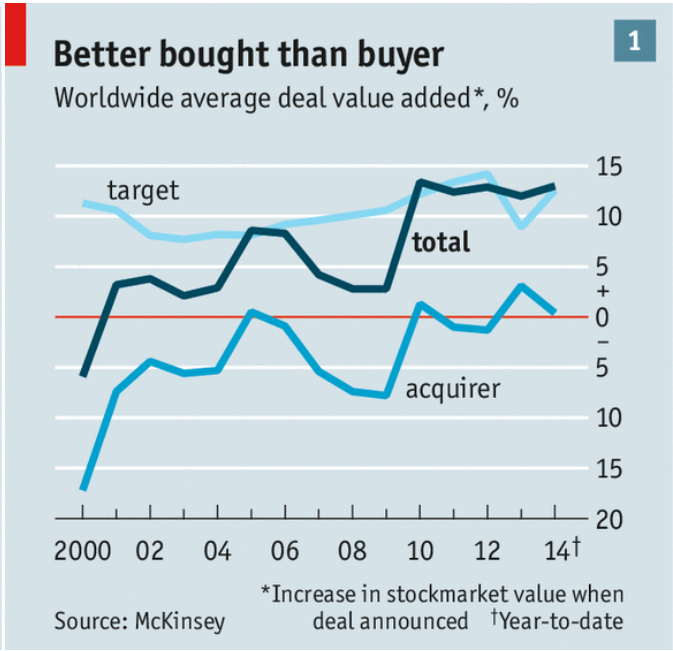
1. M&A creates value overall: average deal generates 1% to 4% gains

2. Provides support to the synergistic theory of mergers

3. One caveat: larger acquisitions are more likely to generate combined losses
   – Moeller, Schlingemann, Stulz (2005)

- Target firm stock price reacts on the announcement of the merger

- Stock price reaction provides a valuable summary of investor opinion about the effect of the merger on firm value
  - Target shareholders earn positive returns
  - Stock price reaction for acquirers is generally negative (or small positive if we focus on the successful deals)



Source:
The Economist, Oct. 7, 2017 Ed.

Long-term performance

- CARs upon merger announcement are affected by short-term fads
  - Stock prices of acquirer and/or target may be “inflated” just prior to merger announcement
    - Synergies are then underestimated
  - Stock prices may revert back to lower level, “once the dust settles”
  - Short-term stock reaction to merger announcement may also reflect changes in opinion about the bidder’s general strategy

- Hard to look over longer horizons (long-run event study over several years) though…
  - … because so many things happen at the same time
  - … correlation is not causation

Does this method provide a clean estimate of the value created by mergers?

No.

Why?

1. Announcing a deal provides information not only about the merger
   - Maybe a signal that target is undervalued?
   - Maybe a signal that acquirer is overvalued?
   - Maybe a signal about change in strategy/industry consolidation/something else
2. Runups
   - Prior to the merger the price may reflect the probability that a merger may take place → Merger premium partly incorporated in pre-announcement price
3. Merger may not occur with 100% probability

What happens after the deal is announced?

- Let's think about the target's stock price
  - Let P_T = the current price of the target's stock
  - Let p = the probability that the deal is completed
  - Let $PV(X)$ = the offer price made by the acquirer
  - Let P_T^{SA} = the price of T as a standalone firm
- Then:

$$P_T = \overbrace{p * \mathrm{PV}(X)}^{\text{Deal Completed}} + \overbrace{(1-p) * P_T^{SA}}^{\text{Deal Not Completed}}, \text{ or}$$

$$P_T = \underbrace{P_T^{SA}}_{\text{Standalone Value}} + p * \underbrace{[\mathrm{PV}(X) - P_T^{SA}]}_{\text{Bid Premium}}.$$

Often p is close to 1

Pfizer to buy Hospira for $90 in cash

Often p is close to 1

Valeant to buy Salix Pharmaceuticals for $173

But not always!

Staples to acquire Office Depot for $11/share

Merger Arbitrage

- Also known as Risk Arbitrage
- Funds that make speculative bets on deal completion

$$P_T = P_T^{SA} + p * [\text{PV}(X) - P_T^{SA}]$$

- Typical strategy, all-cash bid:
  - Borrow money to buy shares in the target
- Typical strategy, all-equity bid:
  - Short acquirer, long target
- Wait until deal closes – (potentially) make profit

Merger Arbitrage

- Also known as Risk Arbitrage
- Funds that make speculative bets on deal completion

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